

Diagnosing Transformation-Order and Inverse-Domain Errors

Answer Key

Precalculus — Advanced function transformations and composition

Quick Answers

- | | |
|------------------------------|--------------------------|
| 1. $x^2 - 6x + 11$ | 5. $2x + 8, -4, 0$ |
| 2. $4(x - 3)^2, 16$ | 6. $(x - 1)^2 + 4, 8$ |
| 3. $2x^2 - 5, 4x^2 + 4x - 2$ | 7. $(x - 3)^2 - 4, x, 6$ |
| 4. $\sqrt{3x - 1}, 2.236$ | 8. — |
-

Curriculum

Level: Precalculus — Advanced function transformations and composition

HSF-BF.A.1c, HSF-BF.B.4 – *problems 3, 4, 6, 7, 8*

HSF-IF.A–HSF-IF.C – *problems 1, 2, 5*

Difficulty 1–4 of 5 across 16 tagged checks.

Common wrong answers

- If they got 3.196: applied f first and g second, computing $g(f(2))$ instead of $f(g(2))$
 - If they got -8: shifted left 8 instead of left 4, so the tested input was -8
 - If they got 0: took the minus branch of the square root, which lands outside the restricted domain
-

1. $f(x) = x^2, g(x) = f(x - 3) + 2$.

Solution: Substituting into f gives $g(x) = (x - 3)^2 + 2$, and expanding: $x^2 - 6x + 9 + 2$.

$x^2 - 6x + 11$

In words: the graph shifts right 3 (an inside operation, so it moves opposite to the sign) and up 2 (an outside operation, which moves the way it reads). This is the one case where inside and outside can be described independently, because there is only one of each.

2. $h(x) = f(2x - 6)$ **with** $f(x) = x^2$.

Solution (a): factor the inside before reading anything off it: $2x - 6 = 2(x - 3)$, so $h(x) = (2(x - 3))^2$.

$4(x - 3)^2$

The horizontal transformations are: compress by a factor of $\frac{1}{2}$ toward the y -axis, then shift right 3. Reading the unfactored inside would have given a shift of 6, which is twice the truth — the compression acts on the shift as well.

Solution (b): $h(5) = (2(5) - 6)^2 = 4^2$.

16

3. $f(x) = 2x + 1$, $g(x) = x^2 - 3$: **both composites.**

Solution: $(f \circ g)(x)$ means g runs first and its output is fed to f :

$$f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1$$

$$2x^2 - 5$$

Reversing the order feeds f 's output to g :

$$g(f(x)) = (2x + 1)^2 - 3 = 4x^2 + 4x + 1 - 3$$

$$4x^2 + 4x - 2$$

They differ because composition is not commutative: the first function's output becomes the second's input, so which one is applied first changes what the second one receives. A single test value makes it concrete — at $x = 1$ the first gives -3 and the second gives 6 .

4. Find and fix: $(f \circ g)(2)$ **with** $f(x) = \sqrt{x + 1}$, $g(x) = 3x - 2$.

The mistake: the notation was read left to right. In $f \circ g$ the function written on the *right* runs first, so g takes the input and f takes g 's output. The student computed $g(f(2))$.

Correct work: compose symbolically first, which makes the order visible:

$$(f \circ g)(x) = f(3x - 2) = \sqrt{(3x - 2) + 1}$$

$$\sqrt{3x - 1}$$

Now substitute: $(f \circ g)(2) = \sqrt{3(2) - 1} = \sqrt{5}$.

$$2.236$$

Watch for: 3.196 , which is $3\sqrt{3} - 2$ — the value of $g(f(2))$. The two composites are different functions, and here they are not even close; a symbolic composition before any arithmetic is what prevents the swap.

5. Find and fix: the graph of $y = f(2x + 8)$.

The mistake: the shift was read off the unfactored inside. A horizontal shift can only be read once the coefficient of x has been factored out, and "shift then compress" also has the order backwards.

Correct work: factor the inside:

$$2x + 8 = 2(x + 4)$$

So $y = f(2(x + 4))$: compress horizontally by a factor of $\frac{1}{2}$, then shift left 4 — not 8 . The compression halves the shift, which is exactly the trap in reading 8 off the constant term.

The inside is zero when $2x + 8 = 0$, that is at $x = -4$, confirming the shift: whatever feature of f sat at input 0 now sits at $x = -4$. Substituting that value back gives $2(-4) + 8 = 0$, as it must.

Watch for: testing $x = -8$ instead. That input gives an inside value of -8 , nowhere near the 0 the check is looking for.

6. $f(x) = \sqrt{x-4} + 1$: **inverse and its domain.**

Solution (a): write $y = \sqrt{x-4} + 1$, swap the variables, and solve for y :

$$x = \sqrt{y-4} + 1 \Rightarrow x-1 = \sqrt{y-4} \Rightarrow (x-1)^2 = y-4$$

Expanded this is $x^2 - 2x + 5$, and in completed-square form

$$(x-1)^2 + 4$$

Solution (b): the domain of f^{-1} is the range of f . Since $\sqrt{x-4} \geq 0$, the smallest output of f is 1, so the range of f is $[1, \infty)$ and the domain of f^{-1} is $x \geq 1$. The algebra alone would happily accept any x , because $(x-1)^2 + 4$ is defined everywhere; the restriction comes from the squaring step, which was only reversible because $x-1$ was known to be non-negative.

Solution (c): $f^{-1}(3) = (3-1)^2 + 4 = 4 + 4$.

8

Check the other way: $f(8) = \sqrt{8-4} + 1 = 2 + 1 = 3$.

7. Find and fix: $f(x) = x^2 - 6x + 5$ on $x \geq 3$.

The mistake: two things at once. A $\pm$ cannot appear in a function — it names two outputs for one input — and "domain: all real numbers" ignores that the domain of an inverse is the range of the original.

Correct work: complete the square to see the vertex:

$$x^2 - 6x + 5 = (x^2 - 6x + 9) - 4$$

$$(x-3)^2 - 4$$

The restriction $x \geq 3$ keeps the right half of the parabola, where f is increasing and one-to-one. Solving $y = (x-3)^2 - 4$ for x gives $x = 3 \pm \sqrt{y+4}$, and the surviving branch is the one whose outputs satisfy $x \geq 3$: the plus. So $f^{-1}(x) = 3 + \sqrt{x+4}$.

The check that this really inverts f is a composition: $((3 + \sqrt{x+4}) - 3)^2 - 4 = (x+4) - 4 = x$.

The range of f on $x \geq 3$ starts at the vertex value -4 and increases, so the range is $[-4, \infty)$ and the domain of f^{-1} is $x \geq -4$ — exactly where $\sqrt{x+4}$ is defined, which is the algebra agreeing with the graph.

Check: $f^{-1}(5) = 3 + \sqrt{5+4} = 3 + 3$.

6

Watch for: 0, the value of the rejected branch $3 - \sqrt{5+4}$. It is a real number and it satisfies $f(0) = 5$, which is why it looks right — but $0 < 3$, so it is outside the restricted domain and is not the inverse's output.

8. Explain: order inside, and the domain of an inverse. (*Open response — flagged for manual review; a model answer follows.*)

Model answer: An expression written inside a function is evaluated before the function is applied, so the transformations it encodes reach the graph in the reverse of the order they are written. In $f(3x - 12)$ the input is multiplied by 3 and then 12 is subtracted, so on the graph the compression by $\frac{1}{3}$ happens first and the shift second. The shift is not 12: factoring gives $3(x - 4)$, so the graph moves right 4. Reading a shift off an unfactored inside is the single most common source of the error, and the sign flips as well, because the input must grow by 4 to produce the value the untransformed function had at 0. Outside operations behave the opposite way — they are applied in the order written and in the direction written — which is why $f(x) + 2$ really does shift the graph up 2.

The domain of f^{-1} is the range of f because the inverse undoes f : its inputs are exactly the outputs f produced, and no others. The algebra usually cannot see this, since the formula obtained by swapping and solving is often defined on a larger set. For $f(x) = \sqrt{x - 4} + 1$ the formula $(x - 1)^2 + 4$ is defined for every real x , but f never outputs anything below 1, so f^{-1} is only defined for $x \geq 1$. When the original function is not one-to-one, a restriction has to be imposed first; that restriction is what makes an inverse exist at all, and it decides which branch of a square root survives.

What a reviewer should look for: the inside factored before a shift is read; inside (input) transformations distinguished from outside (output) ones, including the reversed order and reversed direction; and the statement that a restriction on the original is what makes the inverse a function, with the domain of f^{-1} identified as the range of f .